DO NOW — RECALL

Lesson 083 · Dijkstra: Shortest Paths with Non-negative

Name: _____ Class: _____ Date: _____

1. What does relaxation do?

2. What is a weighted edge?

3. What condition does standard Dijkstra require?

4. Undirected graph edges A–B, A–C, B–D. Give BFS order from A visiting B before C.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Updates a tentative distance via a cheaper path.

2. Model answer

Connection with a cost.

3. Model answer

Non-negative edge weights.

4. Model answer

A,B,C,D.

By the end, you can…

• Dijkstra maintains tentative shortest distances from a start vertex. All edge weights must be non-negative.

• Repeatedly finalise the unsettled node with smallest tentative distance, then relax outgoing edges to improve neighbour estimates.

• Track predecessor nodes alongside distances to reconstruct the shortest route, not just its total cost.

Vocabulary — know these terms

Dijkstra — Non-negative weighted shortest paths.

tentative distance — Current best-known route cost.

relaxation — Replace distance if improved.

predecessor — Previous vertex on best path.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Settle Y and relax Y→Z

After X: Y=4 (predecessor X), Z=10 (predecessor X). Settle Y=4; candidate cost to Z is 4+3=7 , so improve Z to 7 with predecessor Y.

Finish with the path, not just the cost

Settle Z=7. Predecessors: Z←Y←X . Reverse to X→Y→Z , total cost 7. The direct X→Z cost 10 is longer.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Observe

Dijkstra maintains tentative shortest distances from a start vertex. All edge weights must be non-negative.

Apply

Repeatedly finalise the unsettled node with smallest tentative distance, then relax outgoing edges to improve neighbour estimates.

Evaluate

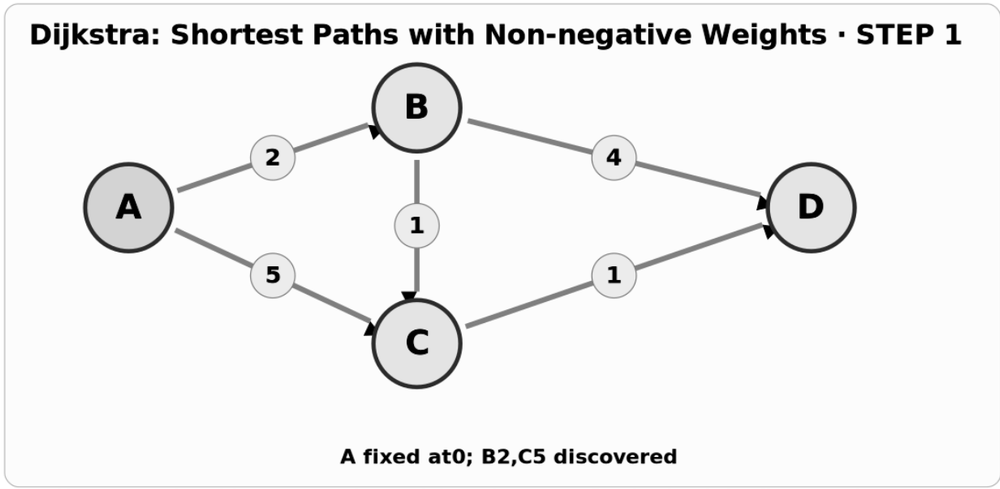
Track predecessor nodes alongside distances to reconstruct the shortest route, not just its total cost.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

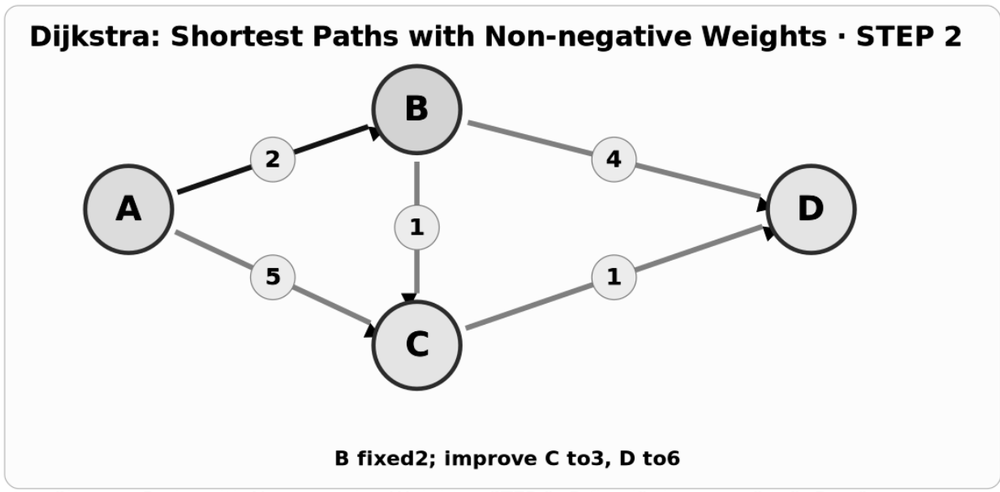
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Dijkstra: Shortest Paths with Non-negative Weights · STEP 1: A fixed at0; B2,C5 discovered

New Learning



Dijkstra: Shortest Paths with Non-negative Weights · STEP 2: B fixed2; improve C to3, D to6

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Dijkstra is guaranteed correct on graphs with arbitrary negative edge costs.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

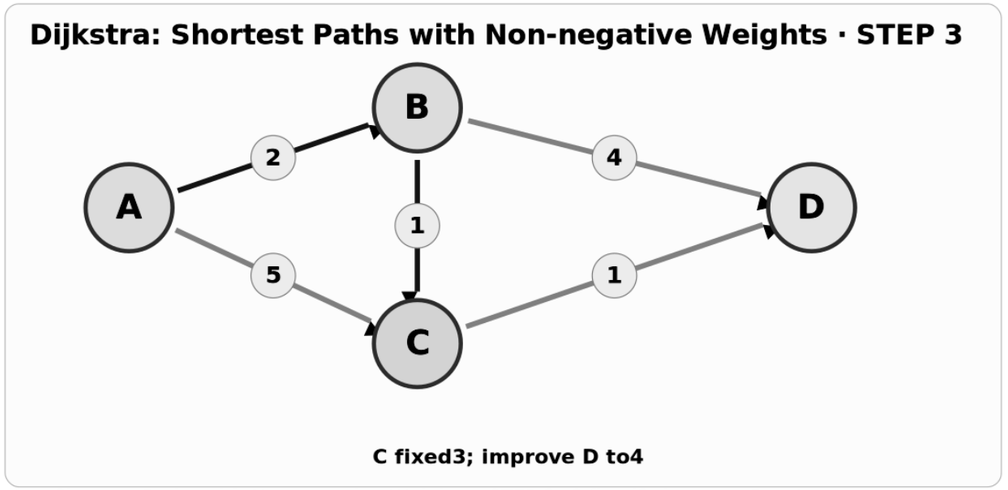
Question 3

Explain why negative edge weights can invalidate Dijkstra's finalisation rule, using a counterexample. [6]

New Learning · original lesson diagrams

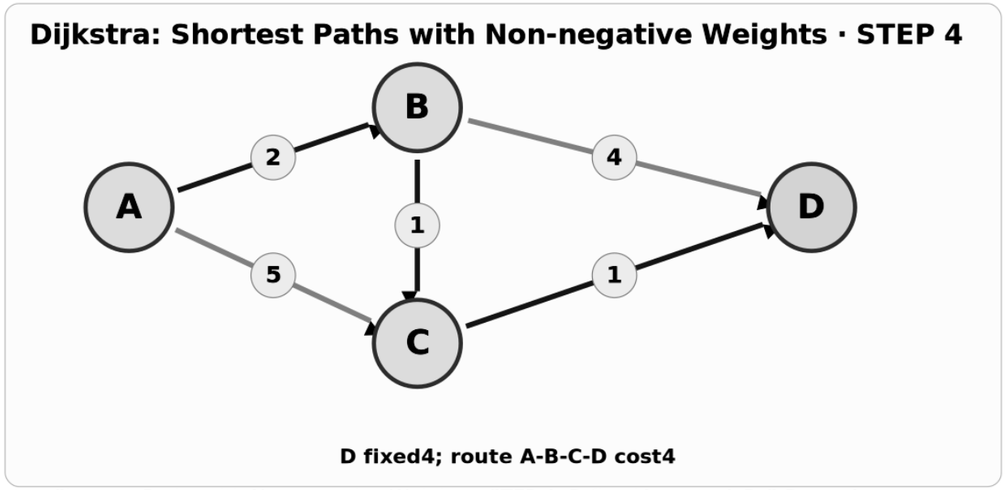
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Dijkstra: Shortest Paths with Non-negative Weights · STEP 3: C fixed3; improve D to4

New Learning



Dijkstra: Shortest Paths with Non-negative Weights · STEP 4: D fixed4; route A-B-C-D cost4

I DO — WORKED EXAMPLE

Follow the worked method

Find the shortest P→R route. Directed edges: P→Q = 3, P→R = 8, Q→R = 2.

Update neighbours

Start with dist(P)=0 . Settle P (smallest unsettled value). Relax P→Q: Q=3 , predecessor Q←P. Relax P→R: R=8 , predecessor R←P. Remaining unsettled distances: Q=3, R=8.

Improve the tentative route

Q=3 is the smallest unsettled distance. Via Q→R, the candidate is 3+2=5 . This beats R=8, so update R=5 and predecessor R←Q. Settled now: P,Q.

Reconstruct the actual path

R=5 is next, and is the target. Follow predecessor pointers R←Q←P ; reverse to obtain P→Q→R , total cost 5 . No vertex with a larger tentative distance was prematurely finalised.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

From A in stated graph, give tentative distances after settling A and B. [6]

Question 2

Continue from C: give final shortest cost and predecessor chain for D. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Settle X; calculate both tentative distances Start at X=0. Which neighbour gets distance 4? Which gets 10? Which unsettled vertex should Dijkstra select next?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

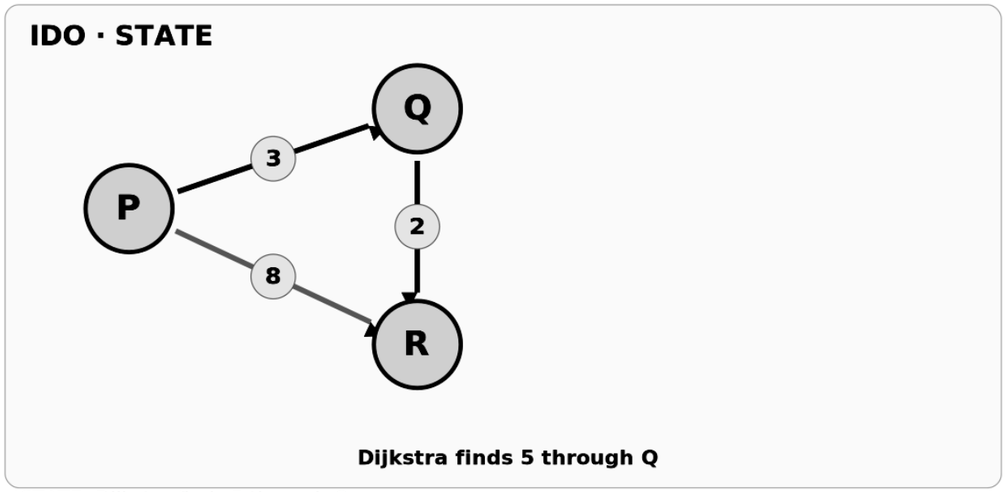
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find the shortest P→R route. Directed edges: P→Q = 3, P→R = 8, Q→R = 2.

Start with dist(P)=0 . Settle P (smallest unsettled value). Relax P→Q: Q=3 , predecessor Q←P. Relax P→R: R=8 , predecessor R←P. Remaining unsettled distances: Q=3, R=8.



IDO · STATE: Dijkstra finds 5 through Q

WE DO — GUIDED PRACTICE

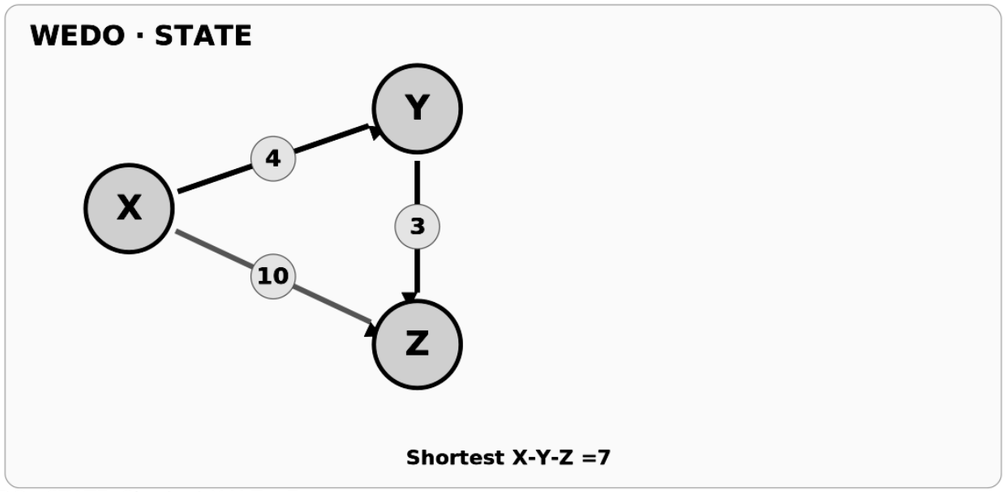
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find shortest X→Z. Directed edges: X→Y = 4, X→Z = 10, Y→Z = 3.

Start at X=0. Which neighbour gets distance 4? Which gets 10? Which unsettled vertex should Dijkstra select next?



WEDO · STATE: Shortest X-Y-Z =7